



Department of Electrical and Electronic Engineering

EEE499P :Final Year Design Project (FYDP)

Weekly Update

Week Number:01

Date: From 12 Oct 2025 To 19 Oct 2025.

ATC Panel:

Chair:Dr. Mohammed Belal Hossain Bhuian

Md. Mahmudul Islam

Sabbir Ahmed

FYDP Group Members:

Shahriar Ahmed

Aban Ahmad

Maisha Mahjabin Nidhi

Daiyan Ibnul Aswad

Week01 Update on: Identification of a real life engineering problem for the proposal.

Topics Assessed:

Topic01: Inspection of Transmission Line

The Research is done in order to find the challenges and possible solutions of transmission line inspection .

Challenges and faults:

The faults in line occur about 2–5 times per 100 km of line per year. The inspectable faults are :

1. The highest fault reason about 40% are due to lightning,rain and humidity which mostly causes rust, breakage of the line or overloading.
2. Equipment failure such as insulators or fittings failure is about 25%.In this case the wire is intact but the fighting is what causes the issues.

Current Solutions (WorldWide):

1. Visual patrols (by foot or vehicle) using binoculars and cameras.
2. Drone-based inspection (UAVs)
3. Smart sensors / IoT devices/Smart Grid
4. Satellite monitoring

Current Solutions (In Bangladesh):

1. Regular visual inspections, manual checks of hardware .
2. Annual checks of vegetation clearance, hardware, insulator & conductor condition.
3. No Smart system is used.
4. Tender Based foreign intervention to supervise and inspect the Transmission Line.
5. No fast or primary inspection methods applied

Summary:

The faults in the transmission line are not rare , worldwide it is assessed with much importance and advanced technologies make sure of quick and efficient assessment regularly .

But in Bangladesh these are yet to be implemented and as a result the power loss is more, the risk of human life remains here and a quick response is impossible. The issues are only assessed annually or when a problem occurs. No precaution or proper maintenance of the lines are done with advanced tech just done by simple on foot and binocular inspection.

Reference:

APA Style

Liu, J., Jia, B., Zhang, Z., Wang, Z., Wang, P., & Geng, J. (2022). Lightning Failure Risk Assessment of Overhead Transmission Lines Based on Modified Dempster–Shafer Theory. *Energies*, 15(7), 2538. <https://doi.org/10.3390/en15072538>

Wang, Li. (2016). The Fault Causes of Overhead Lines in the Distribution Network. MATEC Web of Conferences. 61. 02017. 10.1051/matecconf/20166102017.

In Bangladesh:

Inspection done by Jica last in 2018

https://openjicareport.jica.go.jp/pdf/12303616_01.pdf?utm_source

Tender Based inspection

<https://www.globaltenders.com/tender-detail/dead-line-checking-repairing-work-checking-nut-bolt-tightness-cleaning-disc-insulator-inspection-of-string-and-replace-it-if-required-of-400kv-230kv-132-kv-transmission-lines-under-gopalganj-400230132-kv-grid-substation-as-96511631?>

Topic02: Drone Battery Life Longevity

This study is done for finding the challenges and possible solutions required to extend drone battery life for continuous inspection operations.

Challenges:

1. High power consumption occurs from sensors and processors such as cameras, computing modules and communication systems, which increases battery depletion.
2. Frequent charging or battery replacement from the base station increases downtime and reduces productivity.

Current solutions (worldwide):

1. Wireless Power Transfer from transmission lines or base stations using electromagnetic induction or coupling.
2. Docking systems for battery swapping.
3. Using solar-powered drones.
4. Using brushless motors having high ESCs (Electronic Speed Controllers) and lightweight carbon fiber frames.

Current solutions (Bangladesh):

1. Short flight drones are used for basic aerial photography or surveying.
2. There is no use of self-charging systems in drones yet. Manual charging of batteries is done from the base station.

Summary:

Battery endurance is a major constraint for drones inspecting high-voltage lines. While global industries are advancing toward autonomous, self-charging systems, Bangladesh's drone ecosystem is still developing. Inductive charging from live power lines proposes a practical solution for extended operational time, enabling near-continuous drone operation and aligning with Industry 4.0.

Reference:

1. Mou, X., Gladwin, D., Jiang, J., Li, K., & Yang, Z. (2023). Near-field wireless power transfer technology for unmanned aerial vehicles: A systematical review. *IEEE Journal of Emerging and Selected Topics in Industrial Electronics*, 4(1), 147–158. DOI: 10.1109/JESTIE.2022.3213138
2. Zhu, S. (2025). *The principles of wireless power transfer for drones and optimization of wireless charging efficiency*. *Proceedings of the 2025 International Conference on Applied Energy (ACE)*.
https://www.researchgate.net/publication/391771263_The_Principles_of_Wireless_Power_Transfer_for_Drones_and_Optimization_of_Wireless_Charging_Efficiency